

Data Flow Diagram

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TeamID	SWUID2026221635
ProjectName	A Comprehensive Measure of Well-Being (HDI Prediction & Analysis)
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram(DFD)to show how data moves through your system-between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval/Rounded shape	External Entity	Represents users outside the system boundary. In this project, it includes the User/Analyst (who inputs parameters and views predictions) and the System Admin (who manages datasets).
Rectangle with numbered header	Process	Represents the system operations that transform data. These are: 1. User Authentication (verifies login) 2. Input Validation (checks slider ranges) 3. HDI Prediction Engine (calculates HDI using Linear Regression model) 4. Model Retraining (updates the pickle file).
Rectangle(solid fill, no number)	Data Store	Represents where system data is permanently held. It includes the User Database (credentials), the Model Store (hdi_model.pkl), and Data Storage (hdi_dataset.csv).
Labeled arrow	Data Flow	Shows the pathway and direction of data moving between entities, processes, and stores. Examples include: <i>Login details, Validated Inputs, Predicted HDI Score, and New Training Dataset.</i>

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

